Limited Explanatory Power Between Election Margin of Victory and Demographic Information Using Multiple Linear Regression*

An exploratory investigation on the relationship between a United States congress official and their own characteristics as well as those of their constituents.

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Outcomes of political elections in the United States depend on a multitude of influences, ranging from social, economic, political issues that constituents prioritize. This study compares accuracy of multiple regression models, predicting margin of victory of elections based on demographic information. Subsets of all variables were chosen using variable selection techniques and cross-validated using Root Mean Squared Error (RMSE) for Democrats versus Republican elected officials in Congress. Model results indicated moderate explanatory power, with a limited ability to capture complex patterns in voting data. Future research of prediction using linear regression include incorporation of advanced variable transformations, impactful variable interactions, and Bayesian applications.

1 Introduction

In the United States, there are elections when the two dominant political parties battle with a nail-biting victory and other times where one candidate wins by a landslide. For presidential elects, this margin critically foreshadows their ability to pursue policies (Woolley and Peters (2024)). Voters have a civil duty to assess a candidate's stance on a plethora of social, economical, or political issues and decide who best represents their interests. Past literature, including Kim and Zilinksy's (Silvia Kim and Zilinsky (2024)) Division Does Not Imply Predictability: Demographics Continue to Reveal Little About Voting and Partisanship, indicate a lack of predictive power for voting decisions based on social sorting and mass polarization (2022).

*Project repository available at: https://github.com/christiecngo/math261a_project2.

The authors had used classification methods, including random forest and logistic regression, to predict presidential choice.

However, historically, linear regression methods have not been widely utilized for the purpose of determining whether there will be a close-call election. Due to a greater data size, elected officials in Congress will be considered instead of presidential elections. In this paper, we will construct a multiple linear regression model to assess accuracy of predictions on the margin of victory based on demographics of an elected government official's constituents. Since we will subset the data by political party, the differences between the selected predictors in the best-performing models for Democratic versus Republican parties can be also compared.

2 Data

The dataset is curated from Kimberly Martin's analysis of voting trends on state legislation related to the transgender athletes ban in sports with one bill kept per state, spanning 2020-2023 on Dataverse (2025). There are 29 different states with related bills in total with each row representing individual congressional members. Martin tracks relevant bills based on the Movement Advancement Project (MAP) and the Human Rights Campaign (HRC). Though this dataset is originally intended to study trends specific to the transgender athlete ban in sports, it provides the most reliable and comprehensive demographic characteristics of elected officials. The complete dataset contains 52 columns and 3654 rows in total. There is at least one Democratic and one Republican senator or representative included for each of the 29 states. Key features include binary variables about representatives, such as gender, race, and sexual orientation. Additional information includes the proportions of religious, racial, age, marital status, and educational categories for each official's constituents.

3 Method

3.0.1 Programming Language

This analysis uses R Core Team (2020) Programming software and the following packages: dplyr, MASS, caret, glmnet, and tidymodels.

OpenAI (2025) was also used for guidance extending variable selection to cross validation and generating cleaner figures.

3.0.2 Data Cleaning

Prior to any modeling implementations, variables revealing election outcomes or irrelevant bill information will be omitted. We excluded any observations with a margin of victory equal to 1 as they are uncontested elections and 0 as these are inaccurate entries. We also dropped several rows with any omitted values that were most likely missing at random. After subsetting by party, this resulted in 700 observations for Democrats and 1760 for Republicans. Despite a smaller data size for Democrats, splitting this cohort will best serve our goal of observing prediction accuracy by party.

3.0.3 Full Model and Diagnostics

Through implementing multiple linear regression, our goal is to predict the margin of victory of a congressperson based on their personal demographic information and that of their constituents. The general formula for multiple linear regression is:

$$Y_{nx1} = X_{nxp}\beta_{px1} + \epsilon_{nx1}$$

Y represents the response vector while β is the coefficients associated with the X predictor matrix. ϵ corresponds to the noise in the data due to measurement error or true variations.

3.0.3.1 Response Variable

The response variable is the "margin of victory" column for each elected official's race. The percentage points range from 0.0148% to 99.6887%, with a median of 26.2647%, after removing uncontested elections.

3.0.3.2 Key Predictors

The following predictors remained after screening for variables that contained outcome information and athlete bill-specific columns.

```
 [1] "senate"         "lgbt"           "female"         "black"
 [5] "hispanic"       "nativeamasian"  "dist_votes"     "district_ideal"
 [9] "age25_44"       "age45_64"       "pctover65"      "pctwhite"
[13] "pctblack"       "pcthispanic"    "pctmarried"     "pcths"
[17] "pctbachelordeg" "rural"          "SSMC"           "EP"
```

- Senate (binary 0 or 1) refers to whether the congressperson is in the Senate (1) or House of Representatives (0).

- LGBT, Female, Black, Hispanic, NativeAmAsian are all binary variables that takes on the value 1 if this characteristic describes the official.
- "Dist_votes" is the total number of votes in the official's election while "district_ideal" is the political spectrum of the district on a scale of -1 (Very Liberal) to 0 (moderate) to 1 (Very Conservative).
- Age 25-44, Age 45-64, Over 65 represents the proportion of the district's constituents in that age group.
- [Pct=percentage] White, Black, Hispanic represents the proportion of the district's constituents in each racial group.
- [Pct] Married is the proportion of married individuals in the district
- [Pct] High school, bachelor degree is the proportion of the district who have completed these educational levels.
- Rural represents the proportion of the district considered rural as opposed to urban.
- SSMC is the number of same-sex marriage couples per 1000 households.
- EP is the number of evangelical protestants per 1000 constituents.

The full model includes all predictor variables and serves as a baseline comparison to subsets chosen by variable selection techniques. Data transformations were performed based on diagnostic plots using this comprehensive model.

If our goal had extended to inference with a need for precise confidence intervals and accurate coefficient estimates, the following assumptions of the ordinary least squares linear regression should hold to guaranteed the best linear unbiased estimators (BLUE):

- Linearity: implies that a linear regression model is a good choice for representing the data (response Y versus predictor(s) X)
- Independence of Errors: ensures accurate confidence intervals and p-values based reliable calculations of standard errors.
- Normality of Errors: allows for valid hypothesis tests and confidence intervals constructed
- Equal Variance of Errors: implies that the estimates for the coefficients are the most optimal, as standard errors will be accurate.

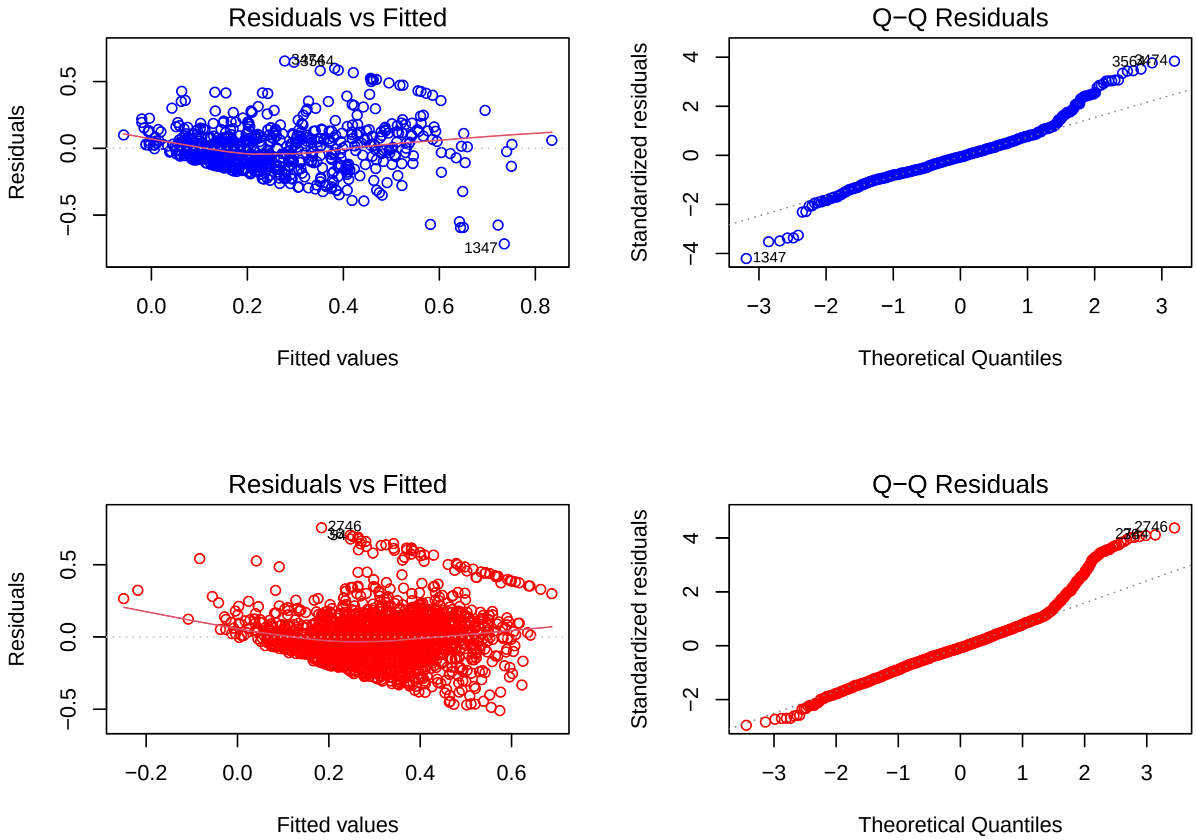


Figure 1: The left column represents the residual versus fitted plot while the right covers the quantile-quantile plot, with blue for Democrat and red for Republican. Primarily in the Q-Q plot, the tail ends show deviation from the dotted line representing a perfect fit with the theoretical distribution. Both subsets violate the homoskedasticity assumption.

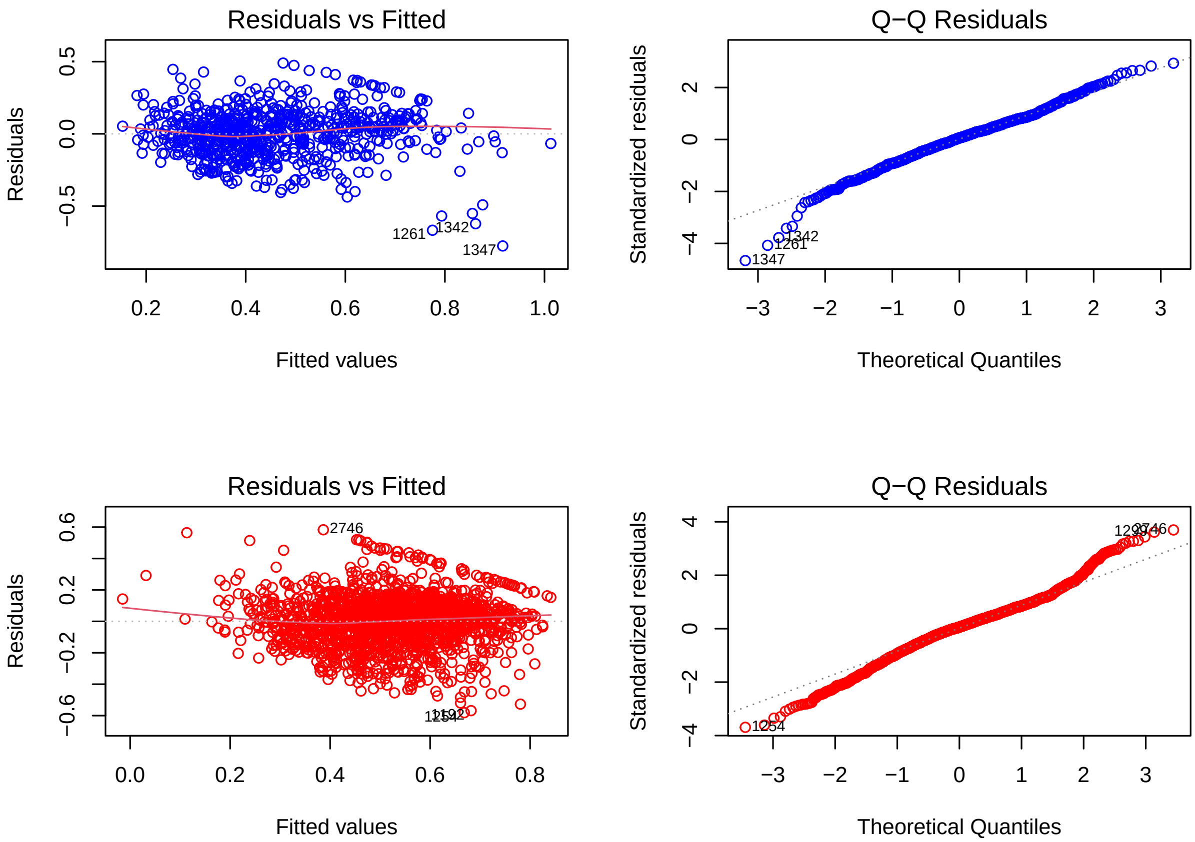
3.0.4 Data Transformations

Due to violation of equal variance, shown clearly in the Q-Q plot of Figure 1, we will first transform the data using Box-Cox transformations. Based on commonly used λ parameters, we find that the square root transformation of the response variable is best for both the Democratic and Republican subsets. This is chosen based on the λ value that minimizes residual variance.

Quick Note: After this transformation, a curve still appeared in diagnostic plots for residuals versus fitted, indicating that there are higher polynomial terms for some predictors for both Republicans and Democrats. We used an iterative loop to plot residuals against each predictor and add a colored LOWESS plot to pinpoint curvature. LOWESS is a method to draw a smoothed line at localized regions. The predictors that showed some curvature for mainly improved for the Democratic subset after including B-Splines terms of degrees of freedom 3. B-Splines is another method for drawing a smooth curve, except using polynomial segments instead.

The Republican subset requires more advanced exploration to resolve linear regression violations, but the analysis will proceed as is. Both subsets also revealed a segment with a linear pattern in the residual versus fitted plot, which could possibly be resolved using piecewise regression or adding interaction terms. We attempted a few pairings for interaction terms but did not see this issue resolve.

Due to the lack of normality in the diagnostic plots, we are unable to construct prediction intervals for point estimates of our test set.



3.0.5 Splitting for Validation

With the limited data size in mind, each subset was split into a training/validation and a testing set with the ratio of 80/20. Cross-validation is performed to select the best model before observing performance on the test set.

3.0.6 Model Comparisons (Variable Selection Techniques)

With the full model as the baseline, we used both LASSO and stepwise sequential search techniques to perform variable selection within each of the 10 folds of our training+validation set. The metric for selection was Root Mean Square Error (RMSE) = $\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$, which is the square root of the mean of differences between the predicted and observed values for the response variable.

For more interpretability without consideration of the scale, we will also observe the R^2 value, which represents the proportion of variance in the data explained by the model.

LASSO is a regularization method that seeks to minimize a penalty term, $\lambda \sum |\beta_j|$. This shrinks the less important estimated coefficients to 0. For stepwise search, we fit a simple linear regression model for each of the $P-1$ predictors and add this predictor to the final model if it exceeds a predetermined threshold.

4 Result

Below are the boxplot comparisons of the results of the RMSE for the full, stepwise selected, and LASSO models:

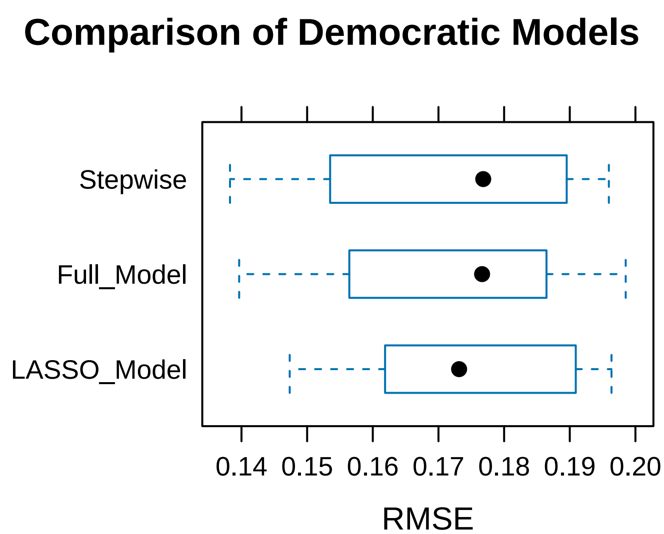


Figure 2: Boxplots for Distributions of Cross-Validation (k=10) RMSE. Democrat (top) and Republican (bottom)

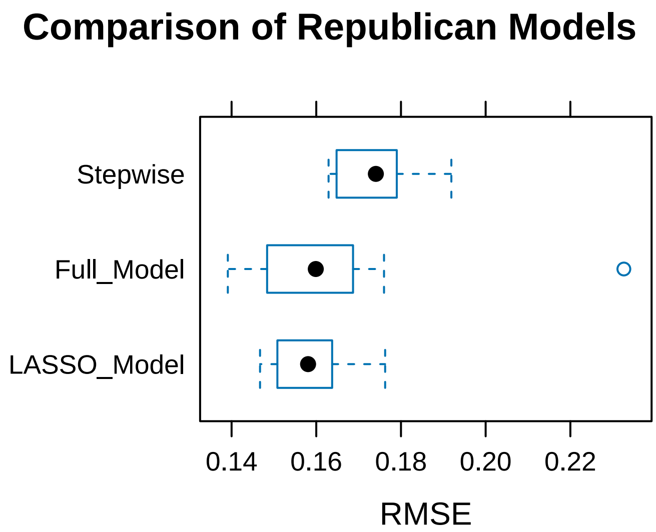


Figure 3: Boxplots for Distributions of Cross-Validation (k=10) RMSE. Democrat (top) and Republican (bottom)

Though each plot in **?@fig-boxplot** shows a wide, overlapping range of RMSE values, we will use the median of each. This is the LASSO model for Democrats and full model for Republicans.

4.0.1 Slope Interpretations

4.0.1.1 Democrat

Variables "zeroed" out by LASSO are considered not important for our response variable.

Key Variables:

- The estimated coefficient for "senate" is 0.01563, representing an expected increase of 0.01563 percentage points in the square root of the margin of victory for senators, when compared to those in the House of Representatives, holding all other variables constant.
- The estimated coefficient for "female" is -0.0148, representing an expected decrease of 0.0148 percentage points in the square root of the margin of victory for female officials, when compared to male, holding all other variables constant.
- The estimated coefficient for "age 45 to 64" is 0.0155, representing an expected increase of 0.0155 percentage points in the square root of the margin of victory given a 1 percentage point increase in constituents age 45 to 64, holding all other variables constant.

- The estimated coefficient for percentage "married" is -0.0743, representing an expected decrease of 0.0743 percentage points in the square root of the margin of victory given a 1 percentage point increase in constituents who are married, holding all other variables constant.
- The estimated coefficient for percentage "white" is -0.0501, representing an expected decrease of 0.0501 percentage points in the square root of the margin of victory given a 1 percentage point increase in constituents who are white, holding all other variables constant. The same interpretations follow for the other race categories.
- The estimated coefficient for percentage "bachelor degree" is 0.0596, representing an expected increase of 0.0596 percentage points in the square root of the margin of victory given a 1 percentage point increase in constituents who have obtained a bachelor degree, holding all other variables constant.

The same interpretations follow for the rest of the variables.

Table 1: Lasso Model Coefficients at Optimal Lambda (0.0013) For Democratic Officials

term	estimate
(Intercept)	0.460028
bs(SSMC, df = 3)1	-0.004904
bs(SSMC, df = 3)2	0.002771
bs(EP, df = 3)1	-0.017841
senate	0.014974
female	-0.014437
black	0.006348
nativeamasian	0.004475
dist_votes	-0.040633
district_ideal	-0.048080
age25_44	0.013370
age45_64	0.013967
pctover65	0.011372
pctwhite	-0.051509
pctblack	0.029235
pcthispanic	0.026049
pctmarried	-0.072528
pcths	-0.017946
pctbachelordeg	0.057034
rural	0.006583
EP	-0.013365

4.0.1.2 Republican

Key Variables:

- The estimated coefficient for “lgbt” is -1.601e-01, representing an expected decrease of 1.601e-01 percentage points in the square root of margin of victory for officials part of the LGBTQ+ community, when compared to those who are not, holding all other variables constant.
- The estimated coefficient for “district ideology” is 1.871e-01, representing an expected increase of 1.871e-01 percentage points in the square root of the margin of victory for senators, given a 1 point increase on the political spectrum (moving towards conservative end), holding all other variables constant.
- The estimated coefficient for “age 45 to 64” is -8.020e-03, representing an expected decrease of 8.020e-03 percentage points in square root of the margin of victory given a 1 percentage point increase in constituents age 45 to 64, holding all other variables constant.
- The estimated coefficient for percentage “married” is 8.943e-03, representing an expected increase of 8.943e-03 percentage points in square root of the margin of victory given a 1 percentage point increase in constituents who are married, holding all other variables constant.
- The estimated coefficient for percentage “white” is 3.370e-03, representing an expected increase of 3.370e-03 percentage points in square root of the margin of victory given a 1 percentage point increase in constituents who are white, holding all other variables constant. The same interpretations follow for the other race categories.

The same interpretations follow for the rest of the variables (note: Evangelical Protestants and same-sex married couples are both singular, which is reasonable due to religious beliefs opposing SSMC; an interaction term was also added for this pair of variables. The proportion with a bachelor degree or in rural living are also singular).

Table 2: Full Model Estimated Coefficients for Republican Officials

	x
(Intercept)	0.3610209
`bs(pctbachelordeg, df = 3)1`	0.0168302
`bs(pctbachelordeg, df = 3)2`	-0.2961542
`bs(pctbachelordeg, df = 3)3`	-0.2351588
`bs(rural, df = 3)1`	-0.0420744
`bs(rural, df = 3)2`	0.0845502
`bs(rural, df = 3)3`	-0.0360679
`bs(SSMC, df = 3)1`	-0.0971241

	x
`bs(SSMC, df = 3)2`	-0.1947846
`bs(SSMC, df = 3)3`	-0.1210683
`bs(EP, df = 3)1`	-0.1121541
`bs(EP, df = 3)2`	0.4153299
`bs(EP, df = 3)3`	-0.1765612
senate	-0.0121716
lgbt	-0.1600701
female	0.0079203
black	0.0178894
hispanic	0.0299283
nativeamasian	-0.0948886
dist_votes	0.0000000
district_ideal	0.1870722
age25_44	-0.0063436
age45_64	-0.0080198
pctover65	-0.0008772
pctwhite	0.0033696
pctblack	-0.0011721
pcthispanic	0.0023048
pctmarried	0.0089431
pcths	-0.0018751
`SSMC:EP`	0.0000912

4.0.2 Testing Results

Table 3: Summary of Best Model Performances on Both Subsets

Subset	Best Model	Validation Median RMSE	Test RMSE	Test R-Squared
Republican	Full	0.1603866	0.1625	0.4517
Democratic	LASSO	0.1651067	0.1757	0.4904

5 Discussion

Ultimately, the best models selected from variable selection techniques yielded moderate explanatory power regarding the prediction of the margin of victory though far from capturing all the true relationships. The method that worked best for Democratic and Republican elected officials were different, with the Democratic model dropping 3 predictors (lgbt status of official, Hispanic race of official, and same sex marriage couple rate of constituents), resulting in 18 distinct variables. It is interesting to see variables associated with more liberal values such as LGBT status and the official being a person of color drop; it is possible that these factors are no longer as impactful when candidates may share the same traits (thus not defining) or the constituents do not consider these statuses positive nor negative.

On the other hand, the model for Republican officials had not officially dropped any variables; 20 distinct variables were included along with the main effect terms for proportion. However, 4 predictors were singular: bachelor degree rate, proportion living in rural areas, rate of same-sex married couples, and rate of Evangelical Protestants. Again, this dependent relationship seems plausible due as conservative religious beliefs are often opposing SSMC; thus, rates of constituents in a same-sex marriage may be under-reported or the area may be deemed unwelcoming.

Democrats had a higher proportion of the variance explained by the multiple linear regression model, with a higher R^2 value at 49.04% but higher RMSE. The Republican model had a slightly lower RMSE but lower R^2 of 45.17% variance explained by the model. This seems reasonable as the Republican cohort has over 1000 more observation points, allowing the model to learn more patterns but perhaps also has greater hidden variation to identify.

5.0.1 Limitations

With a larger subset for both parties, there is a lower risk of overfitting and would have allowed us to implement hyperparameter tuning for a more optimal model. A larger training set would also be beneficial to recognizing complex patterns between the predictors and response. In both cohorts, there were issues with a linear pattern appearing in the residuals versus fitted plot, indicating a missing term (relevant predictor or useful interaction) and preventing accurate construction of prediction intervals. Issues with lack of constant variance was present as well, more notably for the Republican cohort. A more extensive exploration of this political data may resolve these linear violations.

5.0.2 Future Directions

Potential alternative methods could include logit transformations of the predictors, as many are proportions between 0 and 1. For a smaller dataset of this size, Bayesian linear regression could also result in a better model performance and more stable estimates. We may look to separate the effects between representatives in the House and senators as well.

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